A CONNECTED UNRAMIFIED COVER OF A GEOMETRICALLY UNIBRANCH INTEGRAL SCHEME WITH A DOMINANT COMPONENT IS ETALE AND IRREDUCIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10495-10508 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10510-10527, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The author's statement contains a reference environment citing SGA 1, so the card carries a bibliography fragment. That fragment reproduces, field for field and without editing, the @BOOK{SGA1} entry from the Stacks Project snapshot bibliography shipped with this corpus at focused-seeds/AGP/sources/stacks-snapshot-20261011/my.bib (same upstream Stacks Project snapshot). No entry text was altered and no citation was reconstructed from memory.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Unramified morphisms, geometrically unibranch schemes, and the flatness criterion for etaleness; the argument is retained.

Source reference. Original lemma descent-lemma-locally-finite-nr-irred-local-fppf: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10527–10527. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-unramified-dominant-unibranch-is-etale: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10466–10475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-dominant-finite-number-irreducible-components: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10527–10527. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. [1, Expose I, Corollary 9.11] *Let $f : X \to Y$ be a morphism of schemes. Assume*

(1) *Y is integral and geometrically unibranch,*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(2) *at least one irreducible component of X dominates Y,*
(3) *f is unramified, and*
(4) *X is connected.*

Then f is étale and X is irreducible.

Proof. Let $X' \subset X$ be the irreducible component which dominates Y. This means that the generic point of X' maps to the generic point of Y (see for example Morphisms, Lemma morphisms-lemma-dominant-finite-number-irreducible-components). By Lemma lemma-unramified-dominant-unibranch-is-etale we see that f is étale at every point of X'. In particular, the open subscheme $U \subset X$ where f is étale contains X'. Note that every quasi-compact open of U has finitely many irreducible components, see Descent, Lemma descent-lemma-locally-finite-nr-irred-local-fppf. On the other hand since Y is geometrically unibranch and U is étale over Y, the scheme U is geometrically unibranch. In particular, through every point of U there passes at most one irreducible component. A simple topological argument now shows that $X' \subset U$ is both open and closed. Then of course X' is open and closed in X and by connectedness we find $X = U = X'$ as desired. □

References

[1] Grothendieck, Alexander. *Revêtements étales et groupe fondamental (SGA 1)*. Springer-Verlag, Lecture notes in mathematics 224, 1971.